SANJAY HAJABHAI NANDANIYA

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Industry 4.0 & IIoT Architect | Edge AI | IT-OT Transformation Specialist

Building Production-Grade Digital Twins and Industrial Cloud Platforms

A consummate Industry 4.0 and IIoT Architect with over seven years of immersive experience at the confluence of industrial automation, Edge AI, Digital Twins, Data Engineering, ETL and cloud-native IT-OT ecosystems. Distinguished by an ability to translate abstruse industrial challenges into elegant, production-grade architectures that deliver operational clarity, measurable efficiency, and enduring business value across manufacturing, pharma, energy, and automotive domains.

Renowned for intellectual rigor, diagnostic acuity, and executional discipline, he bridges the often-fractured worlds of engineering, operations, and business with uncommon fluency. Combines architectural foresight with hands-on delivery, mentoring teams while driving durable improvements in performance, reliability, and operational transparency. Ideally positioned for senior leadership roles spanning Industry 4.0 architecture, Edge AI engineering, Digital Twin platforms, Data Engineering and large-scale IT-OT transformation programs.

PROFILE SUMMARY

- ✓ Possesses formidable expertise in Edge AI engineering and validation across Qualcomm DSP/NPU, NXP i.MX, and NVIDIA Jetson platforms, with a demonstrable record of deploying resilient LLM, RAG, and computer-vision workloads under the constraints of real-world edge environments.
- ✓ Validated more than 25 AI/ML containers, instituted rigorous inference benchmarking frameworks, enhanced long-run stability, and optimized hardware acceleration across DSP, GPU, and NPU pathways, ensuring predictable performance at scale.
- ✓ Equally adept in the industrial realm, with deep experience in architecting end-to-end IIoT and MES landscapes that unify PLCs, SCADA, HMIs, meters, and historians through OPC UA, BACnet, Sparkplug MQTT, and Kepware.
- ✓ Designed and delivered expansive AWS and Azure–based IoT platforms, orchestrating edge-to-cloud ingestion, microservices, time-series analytics, and Digital Twin models to enable real-time operational intelligence across multi-plant enterprises.
- ✓ Served as a principal technical lead on enterprise-scale initiatives including Azure Digital Twin implementations, cloud historians, and standardized industrial IoT platforms deployed across geographically dispersed facilities, GCP Kubernets.
- ✓ Designed enterprise-grade ETL and Data Engineering pipelines across AWS to power Industrial IoT, Digital Twin, and manufacturing analytics solutions.
- ✓ Brings a mature understanding of regulated manufacturing environments, with practical exposure to 21 CFR Part 11, computer system validation, and equipment qualification frameworks (URS, FRS, IQ, OQ, PQ), ensuring technological ambition remains firmly aligned with compliance and auditability.
- ✓ Recognition:
  - **Nagarro Advantech PM & Lead Appreciation** — Led and executed rigorous RAG-based image validation pipelines, directly improving solution accuracy and contributing to successful customer deployments.
  - **Nagarro Care Award (2×) & NAGP** — Demonstrated consistent ownership and delivery excellence across Edge AI validation initiatives, driving high-quality outcomes and earning repeated organizational recognition.

CORE COMPETENCIES

- Strategic Design and Execution of Industrial 4.0 Ecosystems
- Edge-to-Cloud Systems Orchestration at Industrial Scale
- Production Validation of AI Workloads in Constrained Environments
- Industrial Data Reliability, Lineage, and Trust Engineering
- Translating Manufacturing Operations into Digital Architectures
- Cross-Platform Hardware Acceleration Strategy (DSP, NPU, GPU)
- Operationalizing AI for Safety-Critical and Regulated Plants
- Designing for Long-Run Stability, Resilience, and Recoverability
- Bridging Engineering, Operations, and Business Stakeholders
- Mentorship & Leadership of Cross-Functional Engineering Teams

PROFESSIONAL EXPERIENCE

Nagarro **Since Mar 2024**

Promotional Growth Path:

Staff Engineer	**Dec 2025 – Present**
Associate Staff Engineer (IIoT Cloud & Big Data)	**Mar 2024 – Dec 2025**

PROJECT DETAILS

Project Name:	**Industrial Digital Twin & Smart Manufacturing Platform**
Client:	Global Management Consulting Firm (Fortune 500)
Duration:	Feb-2026 - Present

Performance Highlights:
- Collaborated with solution architects, domain consultants, and customer engineering teams to deliver production-ready Industrial IoT and Digital Twin solutions while supporting cloud modernization, enterprise integrations, and smart manufacturing initiatives.
- Built robust Data Engineering & ETL pipelines using AWS Lambda, AWS Glue, Amazon Redshift, SQL, and cloud-native services for processing, transforming, and analyzing high-volume industrial telemetry.
- Engineered scalable microservices and backend solutions using Python, SQL, TypeScript, FastAPI, Docker, and Kubernetes, optimizing API performance, data validation, and enterprise application scalability.
- Delivered configurable multi-plant portfolio solutions enabling centralized asset management, production monitoring, KPI dashboards, historian integration, alarm management, and operational analytics.
- Designed and implemented Industrial IoT integrations using Ignition, ThingWorx, Kepware, MQTT, Sparkplug B, OPC UA, Modbus TCP/IP, BACnet, REST APIs, and custom protocol plugins for seamless IT/OT connectivity.
- Architected hybrid cloud solutions on AWS, Microsoft Azure, and Google Cloud Platform (GCP) for industrial data ingestion, Digital Twin services, enterprise analytics, and scalable cloud deployments.
- Delivered enterprise-scale Industrial Digital Twin solutions across Life Sciences, Metals, Chemical, Food & Beverage, Building Materials, and Manufacturing industries using cloud-native IIoT architectures.
- Developed responsive React (TypeScript) based Portfolio and Plant Management applications with scalable Python/FastAPI backend APIs for asset hierarchy, Digital Twin visualization, dashboards, and real-time monitoring.

Environment: AWS IoT Core, AWS Lambda, AWS Glue, Amazon Redshift, Azure IoT Hub, Azure Digital Twins, Azure Data Factory, Google Cloud Platform (GCP), Docker, Kubernetes, ETL Pipelines, Ignition, ThingWorx, Kepware, MQTT, Sparkplug B, OPC UA, Modbus TCP/IP, BACnet, REST APIs, React, TypeScript, JavaScript, Python, FastAPI, SQL, PostgreSQL, Industrial Digital Twin, Industrial IoT, Microservices, Portfolio & Plant Management, Manufacturing Analytics

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Project Name: **Edge Computing (LLM, CV, RAG)**
Client: Taiwan Based Leading Edge Device Manufacturer
Duration: May 2025 – Jan 2026

Performance Highlights:
- Delivered production-grade, containerized Edge AI workloads across Qualcomm, NXP, and NVIDIA Jetson platforms with consistent DSP, GPU, and NPU inference behaviour.
- Benchmarked and stabilized LLM and multimodal pipelines including DeepSeek, Llama 3.2, Qwen 2.5, and Tiny Llama across Ollama, LangChain, Genie, FastAPI, and Llama.cpp runtimes.
- Validated and benchmarked hardware acceleration paths across DSP, GPU, and NPU, including CPU fallback modes, using SNPE, QNN, jtop, and platform profiling tools.
- Improved Edge LLM reliability by designing and executing large-document RAG validation pipelines with FAISS, multi-PDF ingestion, grounding verification, and stability analysis.
- Established standardized inference benchmarks covering quantization strategies (Q4, Q8), throughput, cold-boot recovery, and prefill chunking stability.
- Ensured edge system resilience through sustained inference runs, thermal and power compliance testing, CPU and I/O stress testing, and network isolation validation.
- Enabled scalable LLM performance testing by building Locust-based load frameworks with extensive prompt inventories to measure latency and concurrency behaviour.
- Reduced platform debugging time by developing Python-based monitoring and visualization tools for NVIDIA tegrastat and NXP NNStreamer metrics.
- Supported industrial AI adoption through MES PoC validations and cross-partner technical discussions for manufacturing, dairy, and F&B use cases.

Environment: Qualcomm (QCS Platforms), NXP i.MX, NVIDIA Jetson, Ollama, LangChain, Llama.cpp, Genie, Ultralytics (Detection, Segmentation, Classification, Pose Estimation), FastAPI, Python, DSP/NPU/GPU acceleration, Docker, Containerization, FAISS, RAG Pipelines, Locust Load Testing, stress-ng, thermal monitoring, Linux, System Profiling Tools

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Project Name: **IoT Solutions and Structures**
Clients: Biopharma, Energy Grids, Automotive, Building Materials
Duration: September 2024 – April 2025

Performance Highlights:
- Designed and deployed scalable AWS-based IoT architectures enabling real-time, edge-to-cloud data ingestion across multiple industrial domains. Delivered containerized, microservices-based IoT platforms using Docker, eKuiper, RabbitMQ, and Mosquitto to support reliable, low-latency data flow.
- Integrated heterogeneous industrial assets by implementing OPC UA, BACnet, and Sparkplug MQTT for standardized industrial data exchange.
- Built Ignition-based visualization hierarchies with perception modules to enable real-time monitoring of critical industrial equipment.

- Developed ThingsBoard platforms with hierarchical asset models spanning pharma, automotive, energy grid, and parking systems.
- Enabled real-time and historical analytics by implementing Flink-based data processing pipelines with Postgres-backed persistence.
- Strengthened platform security and governance through Linux-based edge hardening and AWS IAM-driven access controls.
- Partnered with cross-functional teams to improve operational visibility, asset utilization, and predictive maintenance outcomes.
- Implemented AWS IoT analytics and digital twin solutions using SiteWise and TwinMaker, enabling contextualized asset modeling, real-time operational insights, and enhanced industrial decision-making.
- Automated cloud infrastructure and CI/CD processes through CloudFormation, CodePipeline, ECS, Elastic Beanstalk, Glue, Redshift, and RDS, ensuring scalable deployments, streamlined data integration, and efficient lifecycle management.

Environment: Greengrass, EC2, Kinesis, ECR/ECS, Postgres, Flink, Fargate, Linux, Greengrass, IAM, S3, ThingsBoard, Docker, Microservices, eKuiper, RabbitMQ, Mosquitto, Python, Kepware, Ignition, Elasticsearch, Logstash, Kibana, Sitewise, TwinMaker, Redshift, Glue, RDS, Beanstalk, Code Pipeline, Cloud Formation

Accenture | Integration Specialist Engineer *(Industry X.0)* **Apr 2023 – Feb 2024**

Project name: **Digital Twin Implementation**
Clients: Oil & Gas, Metal, Pharmaceuticals

Performance Highlights:
- Delivered an enterprise-scale Azure Digital Twin enabling real-time asset visibility and faster operational decisions.
- Integrated live OT data from PLCs, SCADA, and smart devices using OPC frameworks and Azure IoT services.
- Built a scalable AKS-based ingestion and API layer supporting high-volume time-series data.
- Modeled a standardized 11-level asset hierarchy from enterprise to machine level.
- Improved analytics performance through optimized KQL and Azure Data Explorer queries.
- Centralized and stabilized data ingestion using MS-SQL Datalogger and Azure services.
- Accelerated deployments and reliability through Azure DevOps CI/CD pipelines.
- Served as the primary client-facing technical lead, ensuring on-time delivery.

Wipro Technologies Ltd., | Control System Project Engineer *(Domain and IIoT Department)* **Nov 2021 – Apr 2023**

Project name: **Cloud Historian for Big Data Analytics**
Clients: Oil & Gas, Metal
Duration: Apr 2022 – Apr 2023

Performance Highlights:
- Implemented a cloud historian on Azure, consolidating time-series data across five sites.
- Enabled real-time monitoring and predictive insights using Azure Time Series Insights.
- Unified OT and IT data from GE Historian, Aspen, Kepware, and LIMS systems.
- Designed a resilient ingestion architecture using Azure IoT Hub, Stream Analytics, and Data Lake.
- Increased adoption through custom dashboards and notification frameworks.
- Standardized IIoT connectivity across PLCs, SCADA, and metering systems.
- Improved data quality by refactoring and normalizing JSON schemas.
- Led sprint planning and delivery while maintaining daily client alignment.

Project Name: **Industrial IoT Platform – Process in a Box**
Clients: Metal, AC Manufacturer, PSU, Oil & Gas
Duration: Nov 2021 – Apr 2022

Performance Highlights:
- Built and deployed a centralized Industrial IoT and MES platform across six aluminium extrusion plants, standardizing production operations and enabling real-time, data-driven decision-making at scale.
- Enabled plant-wide real-time monitoring and predictive maintenance by integrating PLCs, SCADA systems, energy meters, and smart devices using ThingWorx, Kepware, and OPC-based connectivity.
- Designed and implemented secure, scalable OT–IT architectures, unifying shopfloor and enterprise data through seamless integration of ThingWorx, MSSQL, PostgreSQL, Python services, and federated systems.
- Established a robust data pipeline and data lake integration to improve operational visibility, performance analysis, and cross-plant benchmarking.
- Led infrastructure setup, system migrations, and hybrid networking across on-premise and Azure environments, ensuring reliability, scalability, and high availability. Acted as a primary client-facing engineer, driving technical alignment, stakeholder confidence, and successful delivery across multi-plant deployments.

PREVIOUS EMPLOYMENT HISTORY

Torrent Pharmaceutical Ltd., Gujarat | Executive Engineer **Jun 2019 – Oct 2021**
Operational Project Instrumentation Dept (API, Formulation, SBM, Packing, Utility)

- Led end-to-end MES–L2–L1 integration to optimize manufacturing operations, integrating 100+ machines via Kepware OPC UA, centralized SCADA/historian, Syncade MES, and SQL-based reporting, with workflow and recipe design to improve production control and decision-making.
- Delivered 21 CFR Part 11–compliant upgrades across PLCs, HMIs, SCADAs, and IQMS systems, including audit trails, user access controls, SSRS web reporting, networking, validation (IQ/OQ), and qualification of new and existing equipment across plants.

Sanofi Ltd., Gujarat | Graduate Trainee Engineer **Sep 2018 – Jun 2019**
Instrumentation Dept (API)

TECHNICAL SKILLS

Edge AI & Machine Learning:	Edge AI Validation, LLMs, RAG Systems, Computer Vision
○ Model benchmarking:	Latency, Throughput, Quantization, Memory, Stability
○ Hardware:	NVIDIA Jetson, Qualcomm DSP/NPU, NXP i.MX
○ Runtimes & frameworks:	Ollama, LangChain, Llama.cpp, YOLO
IIoT & Industrial Systems:	IIoT Architecture, IT-OT Integration, Industry 4.0, OPC UA, MQTT Sparkplug, BACnet, Modbus, SCADA, MES, Digital Twins, Industrial Data Pipelines
DevOps & Containerization:	Docker, Kubernetes, Docker Compose, Git, Linux, CI/CD Fundamentals, Containerized Applications
Cloud & Platforms:	AWS, Microsoft Azure, Google Cloud Platform (GCP), AWS IoT Core, AWS Lambda, AWS Glue, Amazon Redshift, Azure IoT Hub, Azure Digital Twins, Azure Data Factory, Azure Data Explorer (Kusto), ETL Pipelines, Cloud-Native & Edge-to-Cloud Architectures
Performance & Validation:	AI Workload Benchmarking and Profiling, Load and Stress Testing, Thermal and Power Analysis, Cross-hardware performance validation
Industrial Automation & MES:	ThingWorx, Ignition, AVEVA, WinCC, Proficy, SCADA–PLC–MES Integrations, Compliance-Driven Industrial Systems
Data & Storage:	SQL and Time-Series Databases, Vector Databases (FAISS), Industrial and Cloud Data Pipelines
Programming & Backend:	Python, FastAPI, Flask, Microservices, Node.js, SQL, React, AI pipelines and backend services, Tomcat, Java
Monitoring & Tooling:	Edge and system profiling tools, Logging and observability stacks

EDUCATION

Bachelor of Technology, Instrumentation and Control, Government Engineering College, Rajkot, Gujarat, India

CERTIFICATIONS & TRAINING

- AWS Fundamentals, AWS Practitioner
- Microsoft Azure AZ-900
- ThingWorx & Kepware 3x Certified
- McKinsey Forward Program
- Ignition
- GenAI APIs for practical applications